

Olympiad Test Paper — Science (Class 7) — Paper 3

Chapter 15: Light

Paper 3 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	15 — Light
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Distractors are seasoned misconceptions: 'concave mirror always magnifies' and 'convex mirror can magnify', a virtual image caught on a screen, plane-mirror images that flip top-bottom instead of left-right, image distance taken as double or half the object distance, convex/concave LENS confused with convex/concave MIRROR, a converging mirror called diverging, a prism said to 'add' or 'create' colours, red bending more than violet, and a single colour split into a spectrum. Do not write on this paper — mark answers on your answer sheet.

1. A pinhole camera made from a box gives an upside-down picture of a candle on its back screen. The single fact about light that best explains the inverted picture is that light:

- (A) bends as it goes through the small hole
- (B) is reflected by the inside walls of the box
- (C) is split into its colours by the hole
- (D) travels in straight lines, so rays from the top of the flame cross through the hole to the bottom of the screen

2. Light from a smooth mirror obeys the law of reflection. When that same beam falls on a rough wall, the wall still scatters light in many directions. In BOTH cases, what is true at each tiny point where a ray meets the surface?

- (A) the angle of incidence is twice the angle of reflection
- (B) the angle of incidence equals the angle of reflection at that point
- (C) the light slows and bends into the surface
- (D) the colour of the light changes on reflection

- 3.** Your face is 60 cm in front of a plane mirror. You see your image, but a friend cannot photograph that image on a piece of card held behind the mirror. The best reason the image cannot land on the card is that the image is:
- (A) real but too dim to expose the card
 - (B) formed 120 cm behind the mirror, beyond the card
 - (C) inverted, so it cancels out on the card
 - (D) virtual — formed where the rays only appear to come from, with no light actually reaching that spot
- 4.** The word CHOICE is written on a card and held up to a plane mirror. Which version below shows correctly how the mirror image reads (letters reversed left-to-right, in reverse order)?
- (A) CHOICE (unchanged)
 - (B) ꞥOIOHC
 - (C) ECIOHC (only the order reversed, letters un-flipped)
 - (D) upside-down CHOICE
- 5.** A boy stands 3 m from a large plane mirror. He then walks 1 m straight toward it. The distance between him and his image is now:
- (A) 4 m
 - (B) 6 m
 - (C) 1 m
 - (D) 2 m
- 6.** A distant streetlamp is viewed in a concave mirror, then a fingertip is brought to within a centimetre of the same mirror. The image of the streetlamp and the image of the fingertip are, respectively:
- (A) real and inverted for the far lamp; virtual, erect and enlarged for the close fingertip
 - (B) both real and inverted
 - (C) both virtual, erect and enlarged
 - (D) virtual diminished for the lamp; real inverted for the fingertip
- 7.** A salesperson claims a particular mirror 'will always show cars behind you larger so they feel closer.' For a convex mirror this claim is:
- (A) correct — a convex mirror magnifies for safety
 - (B) correct only in bright daylight
 - (C) wrong — a convex mirror always forms a diminished image, so cars look smaller and seem farther than they are
 - (D) wrong — a convex mirror inverts the image instead of magnifying it
- 8.** A shopkeeper wants ONE mirror fixed in a high corner so a single camera can watch the entire shop floor. The correct choice and reason are:
- (A) a concave mirror, because it magnifies distant shoppers
 - (B) a plane mirror, because it shows true size
 - (C) a convex mirror, because it gives a wide field of view
 - (D) a concave mirror, because it forms a real image on the camera

9. Two surfaces receive parallel sunlight: a polished steel plate and a sheet of chalk-white paper. You see a sharp bright glare from the steel but only a soft even glow from the paper. The difference is because the paper produces:

- (A) diffuse reflection — its rough surface scatters the rays in many directions
- (B) no reflection at all, only absorption
- (C) refraction that bends the light into the paper
- (D) regular reflection, just like the steel

10. A convex lens held in sunlight can focus the rays to a tiny bright spot hot enough to scorch paper. The property of the lens this demonstrates is that it:

- (A) spreads parallel rays apart
- (B) reflects the sunlight back to a point
- (C) absorbs the heat of the Sun and re-emits it
- (D) converges parallel rays of light to a single point (its focus)

11. Of the four statements about a plane-mirror image, exactly one is FALSE. Which one is FALSE?

- (A) The image is the same size as the object.
- (B) The image is smaller than the object when the object moves farther away.
- (C) The image is laterally inverted.
- (D) The image cannot be formed on a screen.

12. On a misty morning the headlight of a scooter shows a clear straight cone of light cutting through the mist. This is everyday evidence that light:

- (A) travels in straight lines
- (B) bends smoothly around the water droplets
- (C) is a mixture of seven colours
- (D) speeds up as it enters the mist

13. Held up to a plane mirror, the digital number 80 on a clock reads correctly as 08 only if the digits are also reversed in order. Which two-digit display, made of straight segments, would look UNCHANGED in a plane mirror?

- (A) 00
- (B) 80
- (C) 10
- (D) 23

14. A girl 1.6 m tall stands in front of a plane mirror. She steps back so she is twice as far from the mirror as before. The height of her image:

- (A) doubles, because she doubled her distance
- (B) halves, because she is now farther away
- (C) stays 1.6 m — a plane mirror image is always the same size as the object
- (D) becomes 3.2 m, equal to the total object-image distance

15. A torch maker must place a reflector behind the bulb so the torch throws a strong straight beam far down a path. The correct reflector and reason are:

- (A) a convex mirror, because it spreads the light over a wide area
- (B) a plane mirror, because it keeps the beam the same size

- (C) a concave mirror, because it always magnifies the bulb
- (D) a concave mirror, because with the bulb at its focus it sends out a near-parallel beam

16. At a sharp blind bend on a hill road, a large roadside mirror lets drivers see as much of the hidden stretch as possible. The mirror used and the reason are:

- (A) convex, because it gives a wide field of view
- (B) concave, because it magnifies oncoming traffic
- (C) plane, because it shows traffic at true size
- (D) concave, because it forms a real image of the road

17. A teacher lists four image descriptions and asks which CANNOT be produced by any single mirror in this chapter: (i) virtual same-size, (ii) virtual diminished, (iii) real inverted, (iv) real and magnified-erect. Which one is impossible?

- (A) (i) virtual same-size
- (B) (iv) real and magnified-erect
- (C) (ii) virtual diminished
- (D) (iii) real inverted

18. A jeweller uses a hand lens that is thicker in the middle than at its edges to enlarge the upright image of a gemstone held close to it. The lens is a:

- (A) concave (diverging) lens
- (B) convex mirror
- (C) convex (converging) lens
- (D) concave mirror

19. A student writes: 'A polished spoon shows me upside-down on its inner (bowl) side but right-side-up and smaller on its outer (back) side.' Match the two surfaces to their mirror types.

- (A) inner bowl = concave (real inverted image); outer back = convex (erect diminished image)
- (B) inner bowl = convex; outer back = concave
- (C) both surfaces act as plane mirrors
- (D) inner bowl = convex lens; outer back = concave lens

20. A photographer covers a plane mirror with a cloth that has a small clear window. She marks where the image of a candle appears, then tries to project that image onto the cloth. She fails because the plane-mirror image is:

- (A) too small to be projected
- (B) virtual, so no real light actually meets where the image seems to be
- (C) as far in front of the mirror as the candle, not behind it
- (D) inverted, so it cancels itself on the cloth

21. A lens is thinner in the middle than at its edges. Whatever the position of the object, the image it forms is:

- (A) real and inverted
- (B) virtual, erect and magnified
- (C) real and the same size
- (D) virtual, erect and diminished

22. A glass prism throws a band of colours on the wall, while a flat pane of window glass throws no such band. The key reason the prism disperses light but the flat pane does not is that the prism's two refracting faces are:

- (A) coloured, adding colour to the white light
- (B) made of a special rainbow glass
- (C) slanted (not parallel), so the colours, once separated, stay spread apart
- (D) curved like a lens, which creates the colours

23. A digital clock shows 10:01. Seen in a plane mirror, allowing for lateral inversion, the display would appear to read:

- (A) 10:10
- (B) 10:01 (it looks the same)
- (C) 01:01
- (D) 21:51

24. For a concave mirror, as an object is moved from very far away to very close to the mirror, the image changes in which way?

- (A) from virtual erect (far) to real inverted (close)
- (B) from real and inverted (far) to virtual, erect and enlarged (very close)
- (C) it stays real and inverted at every distance
- (D) it stays virtual, erect and diminished at every distance

25. A man walks straight toward a plane mirror at 2 m/s. The speed at which the gap between him and his image shrinks is:

- (A) 2 m/s
- (B) 1 m/s
- (C) the gap does not change
- (D) 4 m/s

26. A convex mirror and a plane mirror are each used as a small make-up mirror to give an enlarged view of the face. Which works, and why?

- (A) the convex mirror, because it magnifies a close face
- (B) the plane mirror, because close objects look bigger in it
- (C) both work equally well for magnifying
- (D) neither — a convex mirror only diminishes and a plane mirror keeps true size, so a concave mirror would be needed instead

27. A laser pointer's dot can be lined up through three small holes in three separate cards only when the holes and the dot lie in one straight line. This experiment is a direct demonstration of:

- (A) the law of reflection
- (B) dispersion of white light
- (C) the straight-line (rectilinear) travel of light
- (D) diffuse reflection from the cards

28. A driver wants a side mirror that BOTH shows a wide stretch of road behind AND keeps every vehicle erect (right-side-up). The best mirror is:

- (A) a concave mirror — wide view and erect image
- (B) a plane mirror — widest possible view

- (C) a convex mirror — it gives a wide field of view and an erect image
- (D) a concave lens fitted as a mirror

29. A magnifying glass held a few centimetres above small print makes the print look bigger and upright, but the SAME lens held far from a window casts a small upside-down picture of the trees outside on a wall. This single lens is a:

- (A) convex lens, which gives an enlarged erect image close up and a real inverted image of distant objects
- (B) concave lens, which can do both kinds of image
- (C) plane mirror used two ways
- (D) convex mirror, which magnifies close and inverts far objects

30. A convex lens forms a sharp, small, upside-down image of a faraway building on a screen. Which describes this image correctly?

- (A) virtual and erect
- (B) virtual and inverted
- (C) real and inverted
- (D) real and erect

31. A pencil is placed 5 cm in front of a plane mirror. The TOTAL distance from the pencil to its image is:

- (A) 5 cm
- (B) 2.5 cm
- (C) 25 cm
- (D) 10 cm

32. White sunlight passed through a prism makes a band whose colours always appear in the same order. Reading from the least-bent to the most-bent colour, the order is:

- (A) violet, indigo, blue, green, yellow, orange, red
- (B) red, yellow, blue (only three colours)
- (C) green, red, violet, blue, orange, yellow, indigo
- (D) red, orange, yellow, green, blue, indigo, violet

33. Two children try to see each other's eyes using a small plane mirror placed between them on a table. They succeed only when the mirror is tilted so that the angle of incidence equals the angle of reflection for the path between their eyes. This is because:

- (A) a mirror reflects light only straight back the way it came
- (B) the mirror bends the light toward whichever child is taller
- (C) light from the eyes is dispersed into colours by the mirror
- (D) for any flat mirror, the reflected ray leaves at the same angle to the normal as the incoming ray

34. A dentist needs a mirror that, held close to a tooth, shows the tooth ENLARGED and the right way up. The correct mirror, and the type of image it gives there, are:

- (A) a convex mirror, giving a magnified erect image
- (B) a plane mirror, giving an enlarged image up close

- (C) a concave mirror, giving a virtual, erect, magnified image
- (D) a concave mirror, giving a real inverted image

35. The capital letters B, C, D, E all share a horizontal line of symmetry but not a vertical one. Held up to an ordinary plane mirror, the letter B would:

- (A) look completely unchanged
- (B) look reversed left-to-right (a backwards B), because the mirror swaps left and right, not top and bottom
- (C) appear upside down
- (D) turn into the letter D

36. A convex (security) mirror in a shop and a convex (rear-view) mirror on a car share which property?

- (A) both can form a real image on a screen
- (B) both always form a virtual, erect, diminished image with a wide field of view
- (C) both magnify nearby objects
- (D) both invert the image left-to-right specially

37. A child is offered three mirrors and asked which one will let them see a much WIDER part of the room than a flat mirror of the same size. The right answer is:

- (A) the concave mirror
- (B) the plane mirror
- (C) no mirror can beat a flat mirror's view
- (D) the convex mirror

38. A student focuses sunlight with a convex lens to a point and burns a dot in a leaf, then replaces it with a concave lens of the same size and cannot burn anything. The concave lens fails because it:

- (A) absorbs all the sunlight before it passes through
- (B) reflects the sunlight back to the Sun
- (C) diverges (spreads) the rays apart instead of bringing them to a point
- (D) splits the sunlight into a harmless rainbow

39. A boy stands 4 m from a plane mirror; his sister stands 1 m from the SAME mirror. Whose image is farther from the mirror, and by how much?

- (A) the sister's image is farther, by 3 m
- (B) the boy's image is farther, by 3 m
- (C) both images are the same distance from the mirror
- (D) the boy's image is farther, by 5 m

40. A concave lens is used in some spectacles. Looking at a printed page through such a lens, the letters appear:

- (A) smaller and upright
- (B) larger and upright
- (C) larger and upside down
- (D) the same size but inverted

41. In a dark hall a single slide projector throws a sharp-edged rectangle of light on the far wall, and a child's hand placed in the beam casts a crisp shadow with the same outline as the hand. Both the sharp rectangle and the crisp shadow are explained by light:

- (A) being absorbed by the air in the hall
- (B) spreading equally in all directions after the lens
- (C) changing colour as it crosses the room
- (D) travelling in straight lines

42. Three uses are listed: (i) a shaving mirror, (ii) a solar cooker's reflector, (iii) a torch reflector. Which mirror type do ALL THREE use?

- (A) convex mirror
- (B) concave mirror
- (C) plane mirror
- (D) a mix: (i) convex, (ii) plane, (iii) concave

43. A prism spreads white light into seven colours; a disc painted with those seven colours, spun very fast, looks whitish. Taken together, these two observations show that:

- (A) the prism creates the seven colours out of white
- (B) white light is made up of the seven colours, which can be split apart and combined back
- (C) white light is a single pure colour
- (D) the spinning disc destroys the colours

44. Standing 2 m from a plane mirror, you read your wristwatch in the mirror; it appears to show 8 o'clock. The actual time on your wrist is:

- (A) 4 o'clock
- (B) 8 o'clock
- (C) 10 o'clock
- (D) 2 o'clock

45. Which single statement about a convex mirror is TRUE for every position of the object?

- (A) the image is virtual, erect and smaller than the object
- (B) the image can be caught on a screen
- (C) the image is larger than the object
- (D) the image is inverted

46. A bus has the words SCHOOL BUS painted in mirror-reversed letters across the front. The reason is so that:

- (A) a driver ahead, glancing in the rear-view mirror, reads SCHOOL BUS the right way round
- (B) it looks decorative on the front
- (C) the letters reflect more sunlight
- (D) the words can be read at night without lights

47. A piece of unpolished aluminium foil, crumpled and then loosely flattened, gives no clear reflection of your face, while smooth shiny foil does. The crumpled foil fails to form an image because its uneven surface causes:

- (A) refraction of the light into the foil
- (B) complete absorption of the light
- (C) diffuse reflection — rays scatter in many directions instead of staying together
- (D) dispersion of the light into colours

48. The capital word FERN is held up to a plane mirror. How many of its four letters would still look like proper (un-reversed) letters, because they are symmetric about a vertical line down the middle?

- (A) all four look unchanged
- (B) only the E looks unchanged
- (C) none of the four — F, E, R and N are all reversed left-to-right
- (D) only the N looks unchanged

49. Comparing a plane, a concave and a convex mirror used for a DISTANT object, which statement is correct?

- (A) all three form real images of a distant object
- (B) only the convex mirror forms a real image
- (C) only the concave mirror can form a real image; the plane and convex always form virtual images
- (D) none of the three can ever form a real image

50. A magnifying glass enlarges print held close; the same lens focuses the Sun to a burning point. Which property links these two abilities?

- (A) the convex lens converges light, bringing rays together
- (B) the lens diverges light, spreading rays apart
- (C) the lens reflects light back to a point
- (D) the lens absorbs and re-emits light as heat

51. A real image of a distant lamp is formed by a convex lens on a screen, but you cannot get any screen image from a magnifying glass held close to a small ant. The difference is that the close ant gives a:

- (A) real image that is simply too small
- (B) real image behind the lens
- (C) coloured image only
- (D) virtual image, which cannot be caught on a screen

52. An object is 7 cm in front of a plane mirror. If the object is moved to 12 cm in front, the image moves:

- (A) 5 cm closer to the mirror
- (B) 10 cm farther behind the mirror
- (C) 5 cm farther behind the mirror
- (D) it does not move

53. A rainbow appears in the sky after rain when the Sun is behind the observer. Each raindrop acts like a tiny:

- (A) mirror, reflecting white light unchanged
- (B) magnifying lens that enlarges the Sun
- (C) filter that lets only one colour through
- (D) prism, splitting sunlight into its colours

54. A laser beam in a smoky room is seen as a perfectly straight bright line. If a flat glass plate is slipped into the beam, the line continues straight on the far side (just shifted). The straightness of the beam across the room is evidence that, in a single uniform medium, light:

- (A) always curves slightly downward
- (B) travels in straight lines
- (C) spreads out into a cone
- (D) separates into seven colours by itself

55. A student says, 'A concave mirror always makes things look bigger.' Give the correct verdict.

- (A) false — a distant object in a concave mirror gives a small, inverted (real) image
- (B) true — for every object position it magnifies
- (C) true — but only for coloured objects
- (D) false — a concave mirror always makes things smaller

56. A convex mirror is chosen over a plane mirror for a delivery van's rear-view mirror mainly because the convex mirror:

- (A) shows everything at exactly true size
- (B) magnifies the cars behind so they look closer
- (C) forms a real image of the road on the glass
- (D) shows a wider area behind the van, even though everything looks a bit smaller

57. If pure GREEN light (a single colour) is shone through a glass prism, what comes out?

- (A) the full seven-colour spectrum
- (B) white light
- (C) only green light, bent a little but not split into other colours
- (D) no light at all

58. Why can you read this page, even though paper is not a light source?

- (A) the paper produces its own light
- (B) the paper reflects room light diffusely, scattering it to your eyes from every part of the page
- (C) the paper forms a regular mirror-image of the words
- (D) the paper refracts light from below the page

59. Held up to a plane mirror, which capital word would still read correctly (unchanged), because every letter is symmetric about a vertical line AND the word reads the same reversed?

- (A) BOOK
- (B) TOOT
- (C) CODE
- (D) MIRROR

60. A teacher pairs each device with the image it forms. Which row is entirely CORRECT?

- (A) plane mirror -> virtual same-size; convex mirror -> virtual diminished; concave lens -> virtual diminished
- (B) plane mirror -> real same-size; convex mirror -> virtual magnified; concave lens -> real
- (C) concave mirror (distant object) -> virtual erect; convex mirror -> real; plane mirror -> inverted
- (D) convex lens -> always virtual; concave mirror -> never real; convex mirror -> magnified

Solutions — Science (Class 7), Chapter 15: Light — Paper 3

Paper 3 — (progressively harder)

Marking: +4 correct, −1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	11	B	21	D	31	D	41	D	51	D
2	B	12	A	22	C	32	D	42	B	52	C
3	D	13	A	23	B	33	D	43	B	53	D
4	B	14	C	24	B	34	C	44	A	54	B
5	A	15	D	25	D	35	B	45	A	55	A
6	A	16	A	26	D	36	B	46	A	56	D
7	C	17	B	27	C	37	D	47	C	57	C
8	C	18	C	28	C	38	C	48	C	58	B
9	A	19	A	29	A	39	B	49	C	59	B
10	D	20	B	30	C	40	A	50	A	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (D) Each ray travels straight; a ray leaving the top of the flame passes through the hole and lands low on the back screen, while a ray from the bottom lands high, so the picture is flipped. The hole does not bend light into a focus, no mirror-like reflection forms the picture, and a tiny hole does not disperse colours. Straight-line travel is the cause.

2. (B) The law of reflection holds at every individual point, even on a rough wall: the surface there has many tiny tilts, so the equal-angle rule still applies locally but the normals point every which way, scattering the light. The angle is not doubled, reflection is not bending into the surface, and reflection does not change colour. Equal angles at each point is the shared truth.

3. (D) A plane mirror always makes a virtual image: the reflected rays diverge and only seem to come from behind the mirror, so no light actually crosses that point and nothing can be captured on a screen. It is not a faint real image, the distance behind equals 60 cm

(not 120 cm) and that is not why it fails, and a plane-mirror image is erect, not inverted. Being virtual is the reason.

4. (B) Lateral inversion swaps left and right, so the whole word is read back-to-front AND each letter is mirror-flipped: C H O I C E becomes E C I O H C with every letter reversed, shown as E C I O H C . The word is not unchanged (it is not vertically symmetric as a whole), reversing only the order without flipping each letter is wrong, and a plane mirror does not turn images upside down — top and bottom are not swapped.

5. (A) The image is always as far behind the mirror as the object is in front. After walking 1 m closer, the boy is $3 - 1 = 2$ m from the mirror, so the image is 2 m behind it. The separation between him and his image is $2 + 2 = 4$ m. 6 m uses his old 3 m position ($3 + 3$); 1 m is just the step he took; 2 m is only his distance to the mirror, not to the image. So 4 m is correct.

6. (A) A concave mirror forms a real, inverted image of a far object (rays converge in front of the mirror), but for an object very close to it the image becomes virtual, erect and magnified — this is why it serves as a shaving mirror. So far = real inverted, close = virtual erect enlarged. It is not 'both' the same, and the lamp/fingertip cases are not swapped. The mixed answer is correct.

7. (C) A convex mirror always gives a virtual, erect, diminished image regardless of object position, so objects look smaller (and thus farther), never larger — that is why warnings say 'objects are closer than they appear.' It does not magnify in any light, and it does not invert (the image is erect). So the magnify claim is wrong because the image is always diminished.

8. (C) A convex mirror spreads reflected rays, so it captures a much wider field of view than a plane or concave mirror of the same size — ideal for surveillance. A concave mirror would magnify and narrow the view, and would not help see the whole floor; a plane mirror shows true size but a narrow view; a real image is irrelevant to coverage. Convex for wide view is correct.

9. (A) The steel is smooth, so parallel rays stay parallel (regular reflection) and form a glare; the paper's microscopically rough surface sends rays off in all directions (diffuse reflection), giving an even glow with no glare and no image. The paper does reflect (you can see it), light is not refracted into it to explain the glow, and it is not regular reflection — that is what the steel does. Diffuse reflection is correct.

10. (D) A convex (converging) lens bends parallel sunlight inward to meet at a focal point, concentrating the energy enough to burn paper. It does not diverge the rays (that is a concave lens), it transmits rather than reflects the light, and it is the concentration of rays — not absorption and re-emission — that scorches the paper. Converging to a point is correct.

11. (B) A plane mirror always gives a same-size image; moving away does NOT shrink it (it only looks smaller to your eye, like any distant thing), so that statement is the false one. The other three are all true: the image is same size, laterally inverted (left-right swapped), and virtual so it cannot be caught on a screen. The 'gets smaller' claim is the false statement.

12. (A) The sharp straight edges of the visible cone show the light is moving in straight lines (the droplets just scatter a little of it toward your eye so you can see the beam). The straight cone is not evidence of bending around droplets, the seven-colour fact is about dispersion not the beam shape, and speed change is not what the straight cone reveals. Straight-line travel is correct.

13. (A) A display looks unchanged in a plane mirror only if it is symmetric about a vertical line AND reads the same after the left-right swap. '00' is made of two identical vertically symmetric zeros, so it stays '00'. '80' reverses to '08'; '10' reverses to '01'; '23' becomes a flipped '32'. Only '00' is unchanged.

14. (C) A plane mirror always forms an image equal in size to the object, no matter how far the object stands; her image stays 1.6 m tall. Distance changes how big it looks to her eye but not the true image size, so it neither doubles nor halves, and 3.2 m confuses size with the object-image separation. The image stays 1.6 m.

15. (D) Placing the bulb at the focus of a concave (converging) mirror makes the reflected rays come out nearly parallel, giving a long strong beam — exactly what a torch or headlight needs. A convex mirror would spread the light and weaken the beam, a plane mirror would not collimate it, and the reason is collimation, not magnifying the bulb. Concave for a parallel beam is correct.

16. (A) A convex mirror's diverging reflection covers a wide field of view, so it shows a long stretch of the hidden road — even though everything looks a little smaller. A concave mirror would magnify but narrow the view (and could invert), a plane mirror shows a narrow strip, and a real image is not the point. Convex for a wide view is correct.

17. (B) A real image formed by a mirror is always inverted, never erect, so 'real and magnified-erect' (iv) cannot occur. The others all happen: a plane mirror gives (i) virtual same-size, a convex mirror gives (ii) virtual diminished, and a concave mirror gives (iii) real inverted for a distant object. So (iv) is the impossible one.

18. (C) A lens thicker in the middle is convex; held close to an object it acts as a magnifying glass, giving an enlarged erect (virtual) image — perfect for a jeweller. A concave lens is thinner in the middle and only diminishes; the two mirror options are not lenses at all (light passes THROUGH a lens). Convex lens is correct.

19. (A) The inner bowl of a spoon curves inward — it is concave, and at normal arm's length it gives a real inverted image, so you appear upside-down. The outer back bulges

out — it is convex, always giving an erect diminished image, so you look small and upright. The types are not reversed, a curved spoon is not a plane mirror, and a spoon reflects (mirror), it is not a lens. Inner concave / outer convex is correct.

20. (B) A plane mirror's image is virtual: the reflected rays diverge and only appear to come from behind the mirror, so no light gathers at the image position and nothing can be projected. The image is same-size (not too small), it lies behind the mirror at the candle's distance (not in front), and it is erect (not inverted). Being virtual is the reason it cannot be projected.

21. (D) A lens thinner in the middle is a concave (diverging) lens; it spreads rays apart and can only form a virtual, erect, diminished image for any object position — it never makes a real or magnified image. So the only correct description is virtual, erect and diminished.

22. (C) In a prism the two faces are inclined to each other, so each colour bends by a slightly different total amount and the colours emerge separated as a spectrum. In flat parallel-faced glass the second face cancels the bending and the colours recombine, so no band shows. The glass adds no colour, needs no special material, and the faces are flat, not curved. Slanted (non-parallel) faces is correct.

23. (B) Lateral inversion reverses the display left-to-right. '10:01' read back-to-front becomes '10:01' again, and each of these digits (1 and 0) is vertically symmetric, so the mirror display still reads 10:01. It is not 10:10 or 01:01 (those mis-order the digits), and 21:51 invents reversed digit shapes incorrectly. The display is unchanged: 10:01.

24. (B) A concave mirror gives a real, inverted image of a distant object, but once the object is brought very close (inside the focus) the image flips to virtual, erect and enlarged — the shaving-mirror effect. So the change is real-inverted to virtual-erect-enlarged. It is not the reverse order, not fixed as real, and 'always virtual diminished' describes a convex mirror, not a concave one.

25. (D) His image approaches the mirror at the same 2 m/s as he does (image distance behind always equals his distance in front). So he closes on the mirror at 2 m/s and the image closes from the other side at 2 m/s, making the man-to-image gap shrink at $2 + 2 = 4$ m/s. 2 m/s ignores the image's motion, 1 m/s halves it, and the gap certainly changes. 4 m/s is correct.

26. (D) A convex mirror always gives a diminished image and a plane mirror always gives a same-size image, so neither can enlarge the face; only a concave mirror held close magnifies (the make-up/shaving mirror). The convex does not magnify, the plane does not enlarge (close objects only fill more of your view), and they do not both work. Neither works — a concave mirror is needed.

27. (C) The beam reaches the far side only when all the holes (and the source and your eye) are collinear, which can only be true if light travels straight. It is not about reflection (the light passes through holes, it is not bounced), not dispersion (a single colour, no spreading into a band), and not diffuse scattering. It demonstrates straight-line travel.

28. (C) A convex mirror gives both a wide field of view and an always-erect (virtual, diminished) image, exactly what a side mirror needs. A concave mirror narrows the view and can invert a distant vehicle; a plane mirror is erect but has a much narrower view; a lens is not a mirror. Convex satisfies both requirements.

29. (A) Only a convex (converging) lens behaves both ways: held close it is a magnifier (enlarged, erect, virtual), and used to image far objects it forms a small real inverted image that can fall on a wall. A concave lens never forms a real image, a plane mirror cannot make a real inverted image of distant trees, and the device transmits light (a lens), so it is not a mirror. Convex lens is correct.

30. (C) Because it lands on a screen, the image is real (light actually meets there), and a real image formed by a convex lens of a distant object is inverted — matching 'small, upside-down.' It cannot be virtual (virtual images never fall on a screen), and a real lens image of this kind is inverted, not erect. Real and inverted is correct.

31. (D) The image sits as far behind the mirror as the object is in front: 5 cm behind. The pencil-to-image distance is therefore $5 + 5 = 10$ cm. 5 cm gives only the object-to-mirror distance, 2.5 cm halves it, and 25 cm squares the number. The total is 10 cm.

32. (D) A prism bends violet the most and red the least, so listing from least-bent to most-bent gives red, orange, yellow, green, blue, indigo, violet (VIBGYOR read backwards). The first wrong option lists most-bent to least-bent, the three-colour option drops four colours, and the last is a jumbled order. Red-to-violet (least to most bent) is correct.

33. (D) The law of reflection (angle of incidence = angle of reflection about the normal) fixes the one tilt that sends light from one child's eyes to the other's. A mirror does not only reflect straight back, it does not preferentially bend toward a taller child, and a plane mirror does not disperse light into colours. The equal-angle law is the reason.

34. (C) Held close (inside its focus) a concave mirror gives a virtual, erect, magnified image — ideal for examining a tooth. A convex mirror only diminishes, a plane mirror keeps true size (never enlarges), and a concave mirror's real image (for far objects) is inverted, which would be useless here. Concave with a virtual erect magnified image is correct.

35. (B) A plane mirror laterally inverts (swaps left and right). B has no vertical symmetry, so it appears as a backwards B. It is not unchanged (only letters symmetric about a VERTICAL axis stay the same), it is not flipped top-to-bottom (mirrors don't swap up-down), and a reversed B is not a normal D. A backwards B is correct.

36. (B) Every convex mirror, whatever its use, forms a virtual, erect, diminished image and offers a wide field of view — so the security mirror and the rear-view mirror share exactly that. Convex mirrors never form a real (screen-catchable) image, never magnify, and do not specially invert. The virtual-erect-diminished-with-wide-view property is shared.

37. (D) A convex mirror's outward curve diverges the reflected rays, taking in a wider field of view than a same-sized plane mirror. A concave mirror actually narrows the field (and magnifies), a plane mirror is the baseline being beaten, and a convex mirror clearly does beat it. The convex mirror gives the widest view.

38. (C) A concave (diverging) lens spreads parallel rays outward, so they never meet at a hot point and cannot scorch the leaf, unlike the converging convex lens. The lens does not absorb all the light (it is transparent), it transmits rather than reflects, and ordinary lens action here is not dispersion into a rainbow. Diverging the rays is the reason it fails.

39. (B) Each image lies as far behind the mirror as its object is in front: the boy's image is 4 m behind, the sister's 1 m behind. The boy's is farther, by $4 - 1 = 3$ m. The sister's is not farther, the images are not equal (the objects differ), and 5 m wrongly adds the two distances instead of subtracting. The boy's image is farther by 3 m.

40. (A) A concave (diverging) lens always forms a virtual, erect, diminished image, so print seen through it looks smaller and the right way up. It cannot enlarge (that is a convex lens close up), it does not invert (the image is erect), and it changes the size (smaller), not leaving it the same. Smaller and upright is correct.

41. (D) Sharp edges on both the projected patch and the shadow show the light moves in straight lines: rays blocked by the hand simply do not reach the wall behind it, giving a crisp outline. Air absorption would dim, not sharpen; rays do not spread in all directions after forming the beam; and no colour change is involved. Straight-line travel explains both.

42. (B) A shaving mirror (enlarged erect image up close), a solar cooker reflector (concentrating sunlight to a hot point), and a torch reflector (collimating into a beam) are all jobs for a concave (converging) mirror. A convex mirror diverges light and cannot concentrate or enlarge; a plane mirror does neither; and they are not a mix. All three use a concave mirror.

43. (B) The prism splits white light into VIBGYOR, and the rapidly spinning disc recombines those colours so the eye sees white again — proving white light is a mixture that can be separated and re-merged. The prism does not create colour (it only separates what is already there), white is not a single pure colour, and the disc combines rather than destroys the colours. White = the seven colours combined.

44. (A) A plane mirror laterally inverts a clock face, and a position that reflects to read T appears at 12 - T on the real clock (for hours; here the mirror image of 4 o'clock looks like 8 o'clock). Since the mirror shows 8, the real time is $12 - 8 = 4$ o'clock. It is not 8 (that is the mirror reading), and 10 or 2 do not satisfy the 12 - T relation. The real time is 4 o'clock.

45. (A) A convex mirror always produces a virtual, erect, diminished image, no matter where the object is. Being virtual, it can never be caught on a screen; it is never larger (always diminished); and it is never inverted (always erect). So 'virtual, erect and smaller' is the universally true statement.

46. (A) Because the rear-view mirror laterally inverts whatever it shows, writing the words already reversed means the mirror flips them back to normal, so the driver ahead reads SCHOOL BUS correctly. It is not for decoration, the mirror-writing does not make letters reflect more sunlight, and lateral inversion has nothing to do with reading in the dark. Correct reading in the mirror is the reason.

47. (C) The microscopically uneven crumpled surface tilts the tiny reflecting facets every which way, so reflected rays scatter (diffuse reflection) and never assemble into an image, unlike smooth foil (regular reflection). Light is reflected, not refracted into the metal; the foil does not absorb it all (you still see brightness); and no colours appear, so it is not dispersion. Diffuse reflection is correct.

48. (C) A capital letter looks unchanged in a plane mirror only if it is symmetric about a VERTICAL line (A, H, I, M, O, T, U, V, W, X, Y). F, E, R and N are not in that set — E and F are symmetric only about a horizontal line, and R and N have no such symmetry — so all four appear reversed. They are not all unchanged, E is not vertically symmetric, and N is not either. None of the four survives unchanged.

49. (C) For a distant object a concave mirror converges the rays to form a real, inverted image, while a plane mirror and a convex mirror always form virtual images. So only the concave one gives a real image. They do not all form real images, it is not the convex one, and it is not 'none' — the concave mirror does. Only-concave-real is correct.

50. (A) Both effects come from the convex lens converging (bending inward) the light: converging the Sun's parallel rays gives a hot focal point, and converging the diverging rays from nearby print produces an enlarged virtual image. The lens does not diverge (that is concave), it transmits not reflects, and the burning comes from concentration, not absorption-and-re-emission. Converging light links both abilities.

51. (D) Held close (inside its focus) a convex lens acts as a magnifier and forms a virtual, erect, enlarged image — virtual images cannot be projected onto a screen, unlike the real image of the far lamp. The close image is not a real one (too small or behind the lens), and it is not merely 'coloured.' It is virtual, so no screen image forms.

52. (C) The image is always as far behind the mirror as the object is in front, so moving the object from 7 cm to 12 cm shifts the image from 7 cm to 12 cm behind — that is 5 cm farther back. It moves farther (not closer), by 5 cm (not 10 cm, which would double it), and it certainly does move. 5 cm farther behind is correct.

53. (D) Sunlight entering a raindrop is dispersed (and internally reflected) much as in a prism, separating white light into the spectrum we see as a rainbow. A plain mirror would reflect white unchanged (no colours), a magnifying lens would not split colours, and a filter removes colours rather than separating them all. Each drop acts like a tiny prism.

54. (B) Within one uniform medium light travels in straight lines, which is why the laser shows a straight track across the room. It does not curve downward, a collimated laser does not spread into a cone over the room, and a single-colour laser does not split into seven colours on its own. Straight-line travel is the evidence.

55. (A) A concave mirror enlarges only when the object is close (inside the focus); for a distant object it forms a small, inverted, real image, so 'always bigger' is false. It does not magnify for every position, the colour of the object is irrelevant, and it does not always shrink either (it CAN magnify up close). 'False — distant object gives a small inverted image' is correct.

56. (D) A convex mirror's wider field of view lets the driver see more lanes behind the van; the trade-off is that everything looks smaller (and farther). A plane mirror shows true size but a narrower view; a convex mirror never magnifies; and it forms a virtual, not a real, image. Wider area (with smaller-looking objects) is the reason.

57. (C) A prism can only separate colours that are already present; a single pure colour like green has nothing to separate, so it merely bends (refracts) and emerges still green. It cannot produce a seven-colour band from one colour, cannot turn one colour into white, and does not block the light entirely. Only bent green light comes out.

58. (B) Paper scatters (diffusely reflects) the room's light in all directions, so light from every part of the page reaches your eyes and you read the contrast of ink and paper. The paper makes no light of its own, it does not form a mirror-like regular reflection (no image), and you do not see it by light passing up through it. Diffuse reflection of room light is why you can read it.

59. (B) For a word to look unchanged it must be a palindrome of vertically symmetric letters. TOOT uses T, O, O, T — all vertically symmetric — and reads the same backwards, so the mirror leaves it as TOOT. BOOK has B and K (not vertically symmetric) and is not a palindrome; CODE has C, D, E (not vertically symmetric); MIRROR has R's and is not a palindrome. TOOT is correct.

60. (A) A plane mirror gives a virtual same-size image, a convex mirror a virtual diminished image, and a concave lens a virtual diminished image — all correct in the first

row. The second row wrongly calls the plane image real and the convex image magnified; the third wrongly makes a distant-object concave image virtual and a convex image real; the fourth wrongly says a convex lens is always virtual and a convex mirror magnifies. The first row is entirely correct.